

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITDSA2-12

Group: 3

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Indicate	Yes	No
Plagiarism report attached		

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Signature 	Date 29 June 2025
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Lecturer's Comments:

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Marks Awarded:	%
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Signature	Date
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Section A

Question 2

2.1)

```
//Create a database called "OLMS"  
use OLMS
```

2.2)

```
//Create each collection with its name  
db.createCollection("Province");  
db.createCollection("Municipality");  
db.createCollection("Facility");  
db.createCollection("Room");  
db.createCollection("Activity");  
db.createCollection("Uses");
```

2.3)

```
//Insert records into the province collection  
db.Province.insertMany(  
  [  
    {  
      "provinceCode": "WC",  
      "provinceName": "Western Cape"  
    },  
    {  
      "provinceCode": "KZN",  
      "provinceName": "KwaZulu-Natal"  
    }  
  ]  
);
```

```
//Insert records into the municipality collection  
db.Municipality.insertMany(  
  [  
    {  
      "munCode": "CPT",  
      "munName": "City of Cape Town",  
      "avgPopulation": 4005016,  
      "provinceCode": "WC"  
    },  
    {  
      "munCode": "ETH",  
      "munName": "eThekweni Metropolitan Municipality",
```

```

        "avgPopulation": 3702231,
        "provinceCode": "KZN"
    }
]
);

//Insert records into the facility collection
db.Facility.insertMany(
[
    {
        "facilityId": "Arts",
        "facilityName": "Artscape Theatre Centre",
        "facilityAddress": "D.F. Malan St, Foreshore, Cape Town, 8001",
        "facilityCapacity": 1500,
        "munCode": "CPT"
    },
    {
        "facilityId": "ICC",
        "facilityName": "Durban ICC",
        "facilityAddress": "45 Bram Fischer Rd, Durban Central, Durban, 4001",
        "facilityCapacity": 10000,
        "munCode": "ETH"
    }
]
);

//Insert records into the room collection
db.Room.insertMany(
[
    {
        "roomNo": "1",
        "roomDescription": "Hall 1",
        "facilityId": "Arts"
    },
    {
        "roomNo": "2",
        "roomDescription": "Hall 2",
        "facilityId": "ICC"
    }
]
);

//Insert records into the activity collection
db.Activity.insertMany(
[
    {
        "activityRef": "MUS01",
        "activityName": "Music Concert"
    },

```

```

        {
            "activityRef": "EXH01",
            "activityName": "Art Exhibition"
        }
    ]
);

//Insert records into the uses collection
db.Uses.insertMany(
[
    {
        "facilityId": "Arts",
        "activityRef": "MUS01",
        "Date": new Date("2025-05-20")
    },
    {
        "facilityId": "ICC",
        "activityRef": "EXH01",
        "Date": new Date("2025-06-15")
    }
]
);

```

2.4)

```

//Search for the Arts facility details in both the facility and room collections
db.Facility.aggregate([
    $match: {facilityId: "Arts"},
    {$lookup: {
        from: "Room",
        localField: "facilityId",
        foreignField: "facilityId",
        as: "rooms"}
    }]).pretty();

```

2.5)

```

//Search for records in the municipality collection that has an average population more than or
equal to 3500000
db.Municipality.find({avgPopulation: {$gte: 3500000}}).pretty();

```

2.6)

```

//Display specific facility details with the corresponding date from both the facility and uses
collection
db.Facility.aggregate([
    $project: {facilityId: 1, facilityName: 1, facilityAddress: 1, facilityCapacity: 1},
    {$lookup: {
        from: "Uses",
        localField: "facilityId",

```

```

    foreignField: "facilityId",
    pipeline: [{ $project: { Date: 1 } }],
    as: "useDate"
  }
}).pretty();

```

2.7)

//Display records from the uses, facility, activity, municipality and province collections that are related to each other

```

db.Uses.aggregate([
  { $lookup: {
    from: "Activity",
    localField: "activityRef",
    foreignField: "activityRef",
    pipeline: [{ $project: { activityName: 1 } }],
    as: "activities" }},
  { $lookup: {
    from: "Facility",
    localField: "facilityId",
    foreignField: "facilityId",
    pipeline: [{ $project: { facilityName: 1, facilityAddress: 1, facilityCapacity: 1, munCode: 1 } }],
    as: "facilities" }},
  { $unwind: "$facilities" },
  { $lookup: {
    from: "Municipality",
    localField: "facilities.munCode",
    foreignField: "munCode",
    pipeline: [{ $project: { munName: 1, provinceCode: 1 } }],
    as: "municipalities" }},
  { $unwind: "$municipalities" },
  { $lookup: {
    from: "Province",
    localField: "municipalities.provinceCode",
    foreignField: "provinceCode",
    pipeline: [{ $project: { provinceName: 1 } }],
    as: "provinces" }},
  { $unwind: "$provinces" }
]).pretty();

```

Question 3

3.1)

```
CREATE DATABASE ONLMS;
```

3.2 a)

//Province nodes

```
CREATE (:Province {provinceCode: 'GP', name: 'Gauteng'});
```

```
CREATE (:Province {provinceCode: 'WC', name: 'Western Cape'});
```

//Municipality nodes

```
CREATE (:Municipality {munCode: 'JHB', name: 'City of Johannesburg', averagePopulation: 4949347, provinceCode: 'GP'});
```

```
CREATE (:Municipality {munCode: 'CPT', name: 'City of Cape Town', averagePopulation: 4005016, provinceCode: 'WC'});
```

//Match provinces and municipalities

```
MATCH (GPT:Province {name: 'Gauteng'}),  
      (WC:Province {name: 'Western Cape'}),  
      (JHB:Municipality {name: 'City of Johannesburg'}),  
      (CPT:Municipality {name: 'City of Cape Town'})
```

```
CREATE (GPT)-[:HAS_MUNICIPALITY]->(JHB),  
      (WC)-[:HAS_MUNICIPALITY]->(CPT);
```



3.2 b)

//Facility nodes

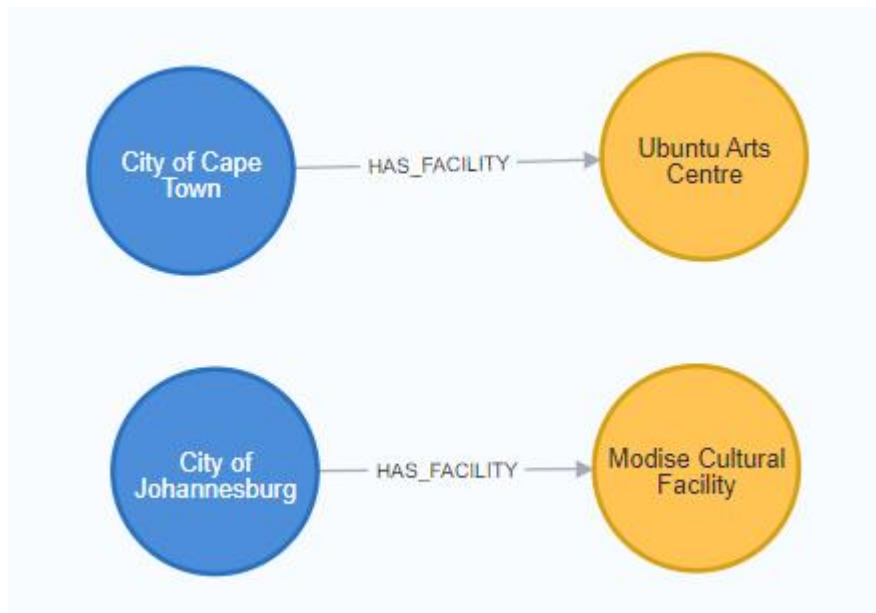
```
CREATE (:Facility {facilityId: 'FAC001', name: 'Modise Cultural Facility', address: '123 Cultural St', capacity: 500, munCode: 'JHB'});
```

```
CREATE (:Facility {facilityId: 'FAC002', name: 'Ubuntu Arts Centre', address: '456 Ubuntu Ave', capacity: 300, munCode: 'CPT'});
```

//Match municipalities and facilities

```
MATCH (JHB:Municipality {name: 'City of Johannesburg'}),  
      (CPT:Municipality {name: 'City of Cape Town'}),  
      (FAC001:Facility {name: 'Modise Cultural Facility'}),  
      (FAC002:Facility {name: 'Ubuntu Arts Centre'})
```

```
CREATE (JHB)-[:HAS_FACILITY]->(FAC001),  
      (CPT)-[:HAS_FACILITY]->(FAC002);
```



3.2 c)

//Room nodes

```
CREATE (:Room {roomNo: 'R001', description: 'Theatre', facilityId: 'FAC001'});
```

```
CREATE (:Room {roomNo: 'R002', description: 'Cinema', facilityId: 'FAC001'});
```

```
CREATE (:Room {roomNo: 'R003', description: 'Exhibition Hall', facilityId: 'FAC002'});
```

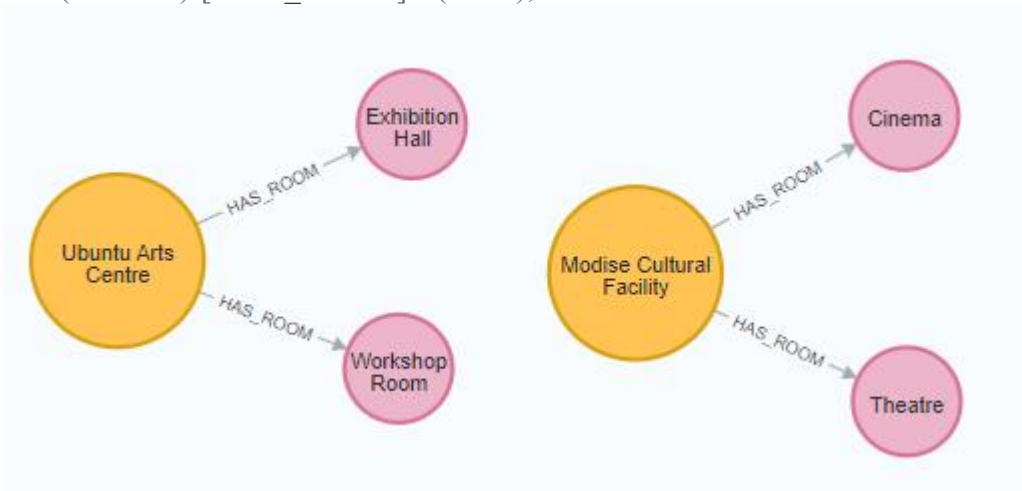
```
CREATE (:Room {roomNo: 'R004', description: 'Workshop Room', facilityId: 'FAC002'});
```

//Match facilities and rooms

```
MATCH (FAC001:Facility {name: 'Modise Cultural Facility'}),  
      (FAC002:Facility {name: 'Ubuntu Arts Centre'}),  
      (R001:Room {description: 'Theatre'}),  
      (R002:Room {description: 'Cinema'}),  
      (R003:Room {description: 'Exhibition Hall'}),  
      (R004:Room {description: 'Workshop Room'})
```

```
CREATE (FAC001)-[:HAS_ROOM]->(R001),  
      (FAC001)-[:HAS_ROOM]->(R002),
```

```
(FAC002)-[:HAS_ROOM]->(R003),
(FAC002)-[:HAS_ROOM]->(R004);
```



3.2 d)

//Uses nodes

```
CREATE (:Uses {facilityId: 'FAC001',activityRef: 'ACT001', Date: date('2025-07-20')});
CREATE (:Uses {facilityId: 'FAC001',activityRef: 'ACT002', Date: date('2025-08-20')});
CREATE (:Uses {facilityId: 'FAC002',activityRef: 'ACT003', Date: date('2025-10-20')});
CREATE (:Uses {facilityId: 'FAC002',activityRef: 'ACT004', Date: date('2025-12-12')});
```

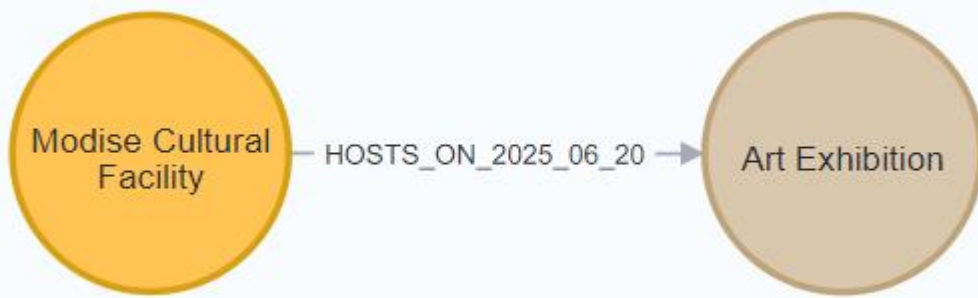
//Activity nodes

```
CREATE (:Activity {activityRef: 'ACT001', name: 'IsiZulu Dance Competition'});
CREATE (:Activity {activityRef: 'ACT002', name: 'Art Exhibition'});
CREATE (:Activity {activityRef: 'ACT003', name: 'Music Concert'});
CREATE (:Activity {activityRef: 'ACT004', name: 'Theatre Performance'});
```

//Match facilities and activity rooms

```
MATCH (FAC001:Facility {name: 'Modise Cultural Facility'}),
      (FAC002:Facility {name: 'Ubuntu Arts Centre'}),
      (ACT001:Activity {name: 'IsiZulu Dance Competition'}),
      (ACT002:Activity {name: 'Art Exhibition'}),
      (ACT003:Activity {name: 'Music Concert'}),
      (ACT004:Activity {name: 'Theatre Performance'})
CREATE (FAC001)-[:HOSTS_ON_2025_07_20]->(ACT001),
      (FAC001)-[:HOSTS_ON_2025_08_20]->(ACT002),
      (FAC002)-[:HOSTS_ON_2025_10_20]->(ACT003),
      (FAC002)-[:HOSTS_ON_2025_12_12]->(ACT004);
```

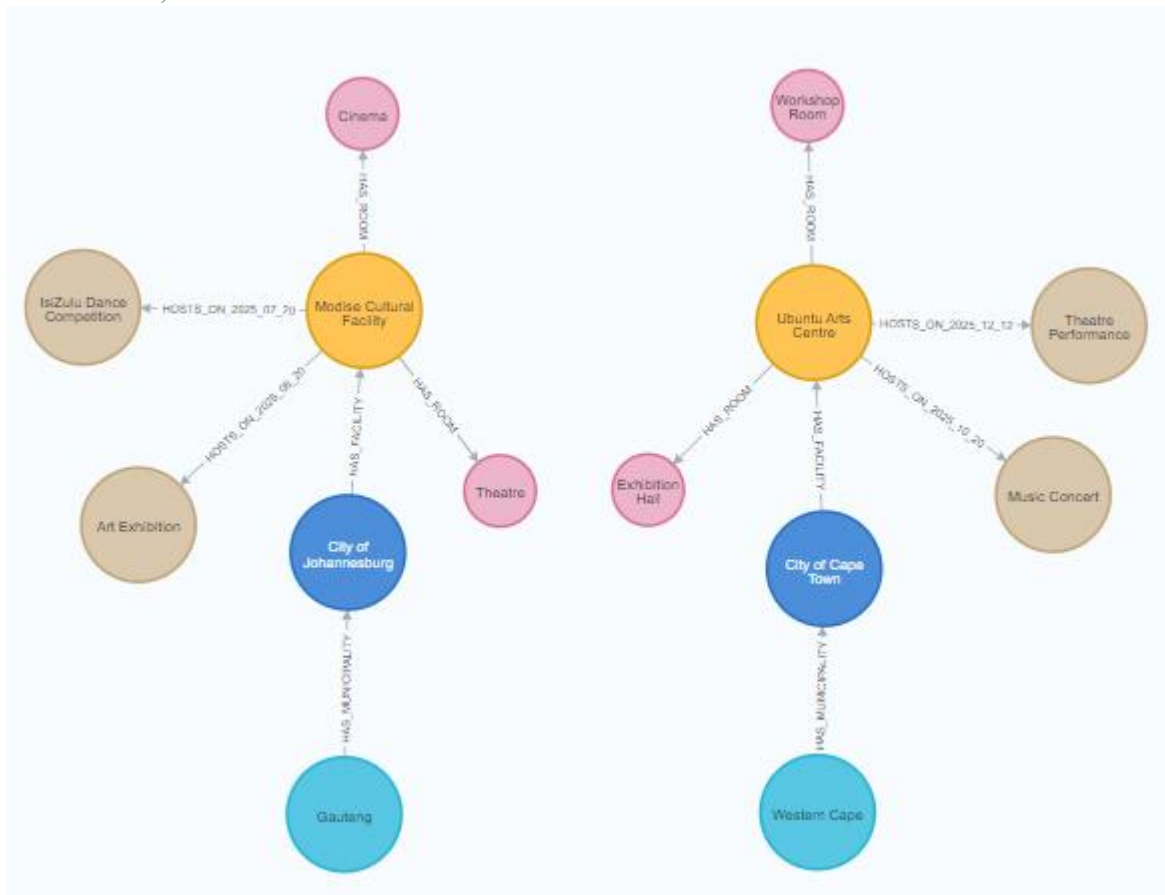




```
//Show all nodes and relationships
```

MATCH(n)

RETURN n;



3.4)

```
//Display the facility capacities in each municipality
```

MATCH (m:Municipality)-[:HAS_FACILITY]->(f:Facility)

```
RETURN m.name AS MunicipalityName,
```

SUM(f.capacity) AS TotalCapacity;

	MunicipalityName	TotalCapacity
1	"City of Johannesburg"	500
2	"City of Cape Town"	300

3.5)

```
//Display the facilities in each municipality in each province
```

MATCH (m:Municipality)-[:HAS_FACILITY]->(f:Facility),

(p:Province)-[:HAS_MUNICIPALITY]->(m:Municipality)

```
RETURN f.name AS FacilityName,
```

m.name AS MunicipalityName,

p.name AS ProvinceName;

Table	FacilityName	MunicipalityName	ProvinceName
Text	"Modise Cultural Facility"	"City of Johannesburg"	"Gauteng"
Code	"Ubuntu Arts Centre"	"City of Cape Town"	"Western Cape"

Table	FacilityName	MunicipalityName	ProvinceName
Text	1 "Modise Cultural Facility"	"City of Johannesburg"	"Gauteng"
Code	2 "Ubuntu Arts Centre"	"City of Cape Town"	"Western Cape"

3.6)

//Display the activity and facility as well as calculate how many days until it occurs from today

MATCH (a:Activity), (f:Facility), (u:Uses)

WHERE a.activityRef = u.activityRef AND

f.facilityId = u.facilityId

RETURN a.name AS ActivityName,

f.name AS FacilityName,

u.Date AS ActivityDate,

duration.inDays(date.realtime(), u.Date).days AS DaysUntilActivity;

Table	ActivityName	FacilityName	ActivityDate	DaysUntilActivity
Text	"IsiZulu Dance Competition"	"Modise Cultural Facility"	"2025-07-20"	21
Warn	"Art Exhibition"	"Modise Cultural Facility"	"2025-08-20"	52
Code	"Music Concert"	"Ubuntu Arts Centre"	"2025-10-20"	113
	"Theatre Performance"	"Ubuntu Arts Centre"	"2025-12-12"	166

	ActivityName	FacilityName	ActivityDate	DaysUntilActivity
1	"IsiZulu Dance Competition"	"Modise Cultural Facility"	"2025-07-20"	21
2	"Art Exhibition"	"Modise Cultural Facility"	"2025-08-20"	52
3	"Music Concert"	"Ubuntu Arts Centre"	"2025-10-20"	113
4	"Theatre Performance"	"Ubuntu Arts Centre"	"2025-12-12"	166

3.7)

//Display all municipalities with an average population more than or equal to 4000000

MATCH (m:Municipality)

WHERE m.averagePopulation >= 4000000

RETURN m.munCode **AS** MunicipalityCode,

m.name **AS** MunicipalityName,

m.averagePopulation **AS** AveragePopulation;

MunicipalityCode	MunicipalityName	AveragePopulation
"JHB"	"City of Johannesburg"	4949347
"CPT"	"City of Cape Town"	4005016

Table

A

Text

Code

	MunicipalityCode	MunicipalityName	AveragePopulation
1	"JHB"	"City of Johannesburg"	4949347
2	"CPT"	"City of Cape Town"	4005016